

Digital Signal Analysis and Processing Exam Notes

CT 704 (BEI/BCT, IV-I) / EX 753 (BEX, IV-II) · TU, IOE · tiers over 45 papers, 2066–2082 BS

2. Z-Transform

4 hrs · 61 questions across the 45 papers · worth about 7 of 80 marks · 3 syllabus sub-topics · the shortest chapter, and the best marks-per-hour in the subject.

Read this first:

- Only two things are ever asked, and they are asked *together*:
 - state the properties of the ROC **TOP 14/45**: a 1 to 5 mark bookwork question, **same answer every year**,
 - find the inverse Z-transform of a given $X(z)$ with a given ROC **TOP 32/45**, for 5 or 6 marks.
- Those two appear as **one** question, Q3, in almost every paper. Learn §2.2 and §2.5 and the chapter is finished.
- The ROC is not decoration.** The same $X(z)$ has a different $x[n]$ for every ROC, and papers exploit that: 81 Ba, 82 Ba and 71 Ch each ask for **three** answers from one $X(z)$.
- Properties with proofs (convolution **HOT 5/45**, time shift, scaling) are worth 3 to 6 marks and come from §2.3.
- Use one method for every inverse:** divide by z , partial-fraction $X(z)/z$, multiply back so every term reads $z/(z-a)$. §2.5 explains why.

2.1 Definition, and how it relates to the DTFT

Define the Z-transform. How is it related to the DTFT? Explain the scaling property **3+3 2**

PIN 3/45 74 Bh, 76 Bh, 73 Ma · the definition is the opening line of most Ch 2 answers

Bilateral (two-sided) Z-transform

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}, \quad z = re^{j\omega}$$

- z is a **complex** variable, so $X(z)$ is defined on a plane, not a line.
- The sum exists only where it converges: that region is the **ROC**, and $X(z)$ **without its ROC is only half an answer**.
- Splitting the sum shows where the two halves live:

$$X(z) = \underbrace{\sum_{n=-\infty}^{-1} x[n] z^{-n}}_{\text{left-sided part}} + \underbrace{\sum_{n=0}^{\infty} x[n] z^{-n}}_{\text{right-sided part}}$$

The right half needs $|z|$ *large*, the left half needs $|z|$ *small*; an ROC is where both hold at once.

Unilateral (one-sided) Z-transform

$$X^+(z) = \sum_{n=0}^{\infty} x[n] z^{-n}$$

- Ignores $n < 0$ entirely, so its ROC is always the outside of a circle and **need not be stated**.
- Importance:** it is the only version that can carry **initial conditions**, so it is what solves a difference equation for a system that is not relaxed.

Relation to the DTFT (quote this)

$$X(e^{j\omega}) = X(z) \Big|_{z=e^{j\omega}}$$

- Put $r = 1$: the Z-transform evaluated **on the unit circle** is the DTFT.
- So the DTFT exists **if and only if the ROC contains the unit circle**. That one sentence answers half the theory questions in this chapter.
- In reverse: $X(z)$ is the DTFT of the exponentially weighted sequence $x[n]r^{-n}$. The r^{-n} is what lets the Z-transform handle growing sequences the DTFT cannot.
- Going round the unit circle once is ω advancing by 2π : the same 2π -periodicity as §1.4.

- Time delay property** of the one-sided transform (73 Ma asks this):

$$Z^+\{x[n-k]\} = z^{-k} X^+(z) + \sum_{m=1}^k x[-m] z^{-(k-m)}$$

- The extra sum is the initial conditions**; the bilateral version has no such term, which is exactly why the one-sided transform is the one that solves a difference equation with given $y[-1]$, $y[-2]$.

Standard pairs (memorise these five)

$x[n]$	$X(z)$	ROC	Note
$\delta[n]$	1	all z	finite duration
$u[n]$	$\frac{1}{1 - z^{-1}} = \frac{z}{z - 1}$	$ z > 1$	pole on the unit circle, so not stable
$a^n u[n]$	$\frac{1}{1 - az^{-1}} = \frac{z}{z - a}$	$ z > a $	right-sided, causal
$-a^n u[-n - 1]$	$\frac{1}{1 - az^{-1}} = \frac{z}{z - a}$	$ z < a $	left-sided, anticausal
$n a^n u[n]$	$\frac{az^{-1}}{(1 - az^{-1})^2}$	$ z > a $	repeated pole; also $\frac{az}{(z - a)^2}$

Rows 3 and 4 have the identical $X(z)$ and differ only in the ROC. That is the whole reason the ROC must be given, and the reason an inverse cannot be started until you have read it.

2.2 Region of convergence

Define the ROC. State and explain the properties of the region of convergence 1 2 3 4 5

TOP 14/45 80 Bh, 73 Ch, 72 Ch, 81 Bh, 77 Ch, 71 Bh, 70 Bh, 79 Ba, 72 Ka, 71 Shr, 73 Bh, 78 Ch, 70 Ma, 74 Ma · **pure bookwork, and it is asked every single year**

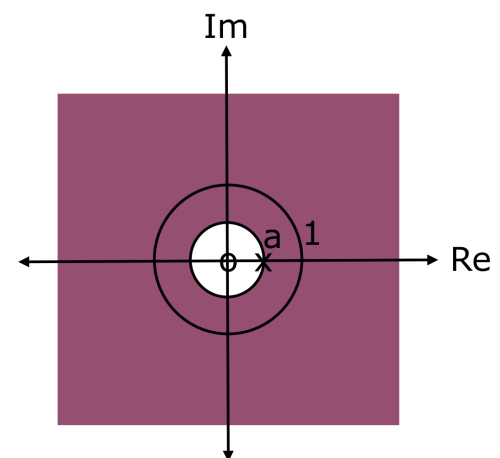
Definition

- The **ROC** is the set of values of z for which $\sum_n |x[n]| |z|^{-n}$ converges, i.e. where $X(z)$ actually exists.
- It depends only on $|z| = r$, never on the angle, which is why it is always drawn as circles.

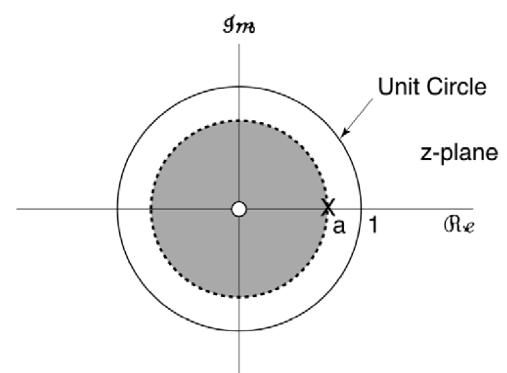
The eight properties (this is the answer, in this order)

1. The ROC is a **ring or a disc** in the z -plane, centred on the origin.
2. The **DTFT** of $x[n]$ converges absolutely if and only if the ROC includes the unit circle.
3. **The ROC cannot contain any pole.** $X(z)$ is infinite at a pole, so the sum cannot converge there.
4. If $x[n]$ is **finite duration**, the ROC is the entire z -plane, possibly except $z = 0$ and/or $z = \infty$.
5. If $x[n]$ is **right-sided** ($x[n] = 0$ for $n < N_1$), the ROC extends **outward** from the outermost finite pole, possibly including $z = \infty$.
6. If $x[n]$ is **left-sided** ($x[n] = 0$ for $n > N_2$), the ROC extends **inward** from the innermost non-zero pole, possibly including $z = 0$.
7. If $x[n]$ is **two-sided**, the ROC is a **ring** bounded inside and outside by a pole, containing none.
8. The ROC must be a **connected region**.

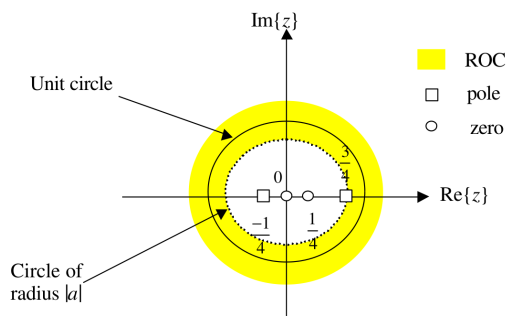
For a 4 or 5 mark answer add one line of *why* to properties 3, 5 and 6, and sketch the three cases. For 1 or 2 marks, the definition plus properties 1, 3 and 5 is enough.



Right-sided $x[n] = a^n u[n]$: the series $\sum (az^{-1})^n$ converges for $|az^{-1}| < 1$, i.e. $|z| > |a|$: everything **outside** the pole circle. BB Sir, deck 2.1



Left-sided $x[n] = -a^n u[-n - 1]$: same $X(z) = z/(z - a)$, but the ROC is $|z| < |a|$, the disc **inside** the pole circle. BB Sir, deck 2.1



Two-sided: the ROC is the ring between two poles. It contains the unit circle here, so this signal has a DTFT. Chapterwise Ch3

Stability and causality from the ROC (asked with the properties)

- **Causal** $\Leftrightarrow h[n] = 0$ for $n < 0 \Leftrightarrow$ the ROC is the **outside of the outermost pole**, and includes $z = \infty$.
- **BIBO stable** $\Leftrightarrow \sum |h[n]| < \infty \Leftrightarrow$ the **ROC contains the unit circle**.
- **Causal and stable together** $\Leftrightarrow$ **every pole lies strictly inside the unit circle**. This is the single most quoted line in Chapters 2, 3 and 6.
- A stable but non-causal system is possible: the ROC is a ring that contains the unit circle but does not reach $z = \infty$.

Reading the ROC off a question (the skill the numericals need)

1. Find the poles.
2. Mark their radii on a line.
3. The given ROC lies in one of the gaps. Any pole with radius below that gap is **right-sided**; any pole above it is **left-sided**.

A given ROC that contains a pole is impossible: two papers print exactly that, see Ch2-num §4.

2.3 Properties of the Z-transform

State and prove the convolution property; describe causality and stability in terms of the ROC 2+4

3+3 1+3+3

HOT 5/45 76 Ash, 72 Ka, 70 Asa, 72 Ma, 72 Ash · **the most-asked property by far**

Property	Transform	ROC
Linearity	$a x_1[n] + b x_2[n] \rightarrow a X_1(z) + b X_2(z)$	at least $R_{x_1} \cap R_{x_2}$
Time shift	$x[n - k] \rightarrow z^{-k} X(z)$	R_x , except possibly $z = 0$ or $z = \infty$
Scaling in z (multiply by a^n)	$a^n x[n] \rightarrow X(a^{-1}z)$	$ a r_1 < z < a r_2$
Differentiation	$n x[n] \rightarrow -z \frac{dX(z)}{dz}$	R_x
Time reversal	$x[-n] \rightarrow X(z^{-1})$	$\frac{1}{r_2} < z < \frac{1}{r_1}$
Convolution	$x_1[n] * x_2[n] \rightarrow X_1(z) X_2(z)$	at least $R_{x_1} \cap R_{x_2}$
Multiplication	$x_1[n] x_2[n] \rightarrow \frac{1}{2\pi j} \oint_C X_1(v) X_2\left(\frac{z}{v}\right) v^{-1} dv$	at least $r_{1_1} r_{1_2} < z < r_{2_1} r_{2_2}$

Convolution, proved (the 3 to 6 mark answer)

- Start from the definition of the convolution sum:
 $x[n] = \sum_k x_1[k] x_2[n - k]$.
- Transform both sides:

$$X(z) = \sum_n \left[\sum_k x_1[k] x_2[n - k] \right] z^{-n}$$

- **Interchange the order of summation** (the step that earns the mark):

$$X(z) = \sum_k x_1[k] \sum_n x_2[n - k] z^{-n}$$

- The inner sum is the **time-shift property**:
 $\sum_n x_2[n - k] z^{-n} = z^{-k} X_2(z)$.

$$X(z) = X_2(z) \sum_k x_1[k] z^{-k} = X_1(z) X_2(z)$$

- **"At least" the intersection:** a pole of one can be cancelled by a zero of the other, which makes the ROC larger than the intersection. Say this: it is the part most answers miss.

Time shift, proved 3 69 Ch

- Let $y[n] = x[n - k]$, so $Y(z) = \sum_n x[n - k] z^{-n}$.
- Substitute $m = n - k$, so $n = m + k$:

$$Y(z) = \sum_m x[m] z^{-(m+k)} = z^{-k} \sum_m x[m] z^{-m} = z^{-k} X(z)$$

- **Why the ROC can change:** z^{-k} adds poles at $z = 0$ (for $k > 0$) or at $z = \infty$ (for $k < 0$), so those two points may drop out.

Scaling, proved 3 74 Bh

$$Z\{a^n x[n]\} = \sum_n a^n x[n] z^{-n} = \sum_n x[n] (a^{-1} z)^{-n} = X(a^{-1} z)$$

- The ROC $r_1 < |z| < r_2$ becomes $r_1 < |a^{-1} z| < r_2$, i.e. $|a| r_1 < |z| < |a| r_2$: **every pole is scaled by a** .
- Example: $u[n] \rightarrow \frac{1}{1-z^{-1}}$, so $a^n u[n] \rightarrow \frac{1}{1-az^{-1}}$, pole moved from 1 to a .

Differentiation, used in every na^n question

- Differentiate $X(z) = \sum x[n] z^{-n}$ term by term and multiply by $-z$:

$$Z\{n x[n]\} = -z \frac{dX(z)}{dz}$$

- Applied to $a^n u[n]$ it gives $Z\{na^n u[n]\} = \frac{az^{-1}}{(1-az^{-1})^2}$, $|z| > |a|$: **the answer 78 Ch and 70 Ma both ask for**.

2.4 Finding $X(z)$ and locating the ROC**Find the Z-transform of the given signal and locate its ROC 2+4 3+6 4+6**

HOT 6/45 70 Ch, 70 Asa, 75 Ash, 75 Ch, 75 Bh, 73 Ma · worked in Ch2-num §6

Method

1. Split $x[n]$ into its terms.
2. Transform each with a standard pair, **writing its own ROC beside it**.
 - $a^n u[n] \rightarrow \frac{1}{1-az^{-1}}$, ROC $|z| > |a|$ (outside).
 - $-b^n u[-n-1] \rightarrow \frac{1}{1-bz^{-1}}$, ROC $|z| < |b|$ (inside).
 - **Watch the sign:** a $+b^n u[-n-1]$ term transforms to $-\frac{1}{1-bz^{-1}}$.
3. The ROC of the sum is the **intersection** of the individual ROCs.
4. If the intersection is empty, **the Z-transform does not exist**: and that is the full answer. 70 Ch prints exactly such a signal (Ch2-num §6).

Two standard transforms the papers ask for by name

Why the intersection can be empty

- A right-sided term needs $|z| > R_1$; a left-sided term needs $|z| < R_2$.
- The ring $R_1 < |z| < R_2$ exists **only if $R_1 < R_2$** : the right-sided pole must be the *inner* one.
- $R_1 = R_2$ or $R_1 > R_2$ leaves nothing, so $X(z)$ does not exist even though each term separately transforms perfectly well.

- $x[n] = na^n u[n]$ (78 Ch, 70 Ma): use differentiation:

$$X(z) = \frac{az^{-1}}{(1 - az^{-1})^2}, \quad |z| > |a|$$

- Derivation in one line:

$$-z \frac{d}{dz} \frac{z}{z-a} = -z \cdot \frac{(z-a) - z}{(z-a)^2} = \frac{az}{(z-a)^2}.$$

- $x[n] = \cos \omega n$ for $n \geq 0$ (74 Ma): write the cosine as two exponentials and add two geometric series:

$$X(z) = \frac{1 - z^{-1} \cos \omega}{1 - 2z^{-1} \cos \omega + z^{-2}}, \quad |z| > 1$$

Its two poles are $e^{\pm j\omega}$, on the unit circle, which is why the ROC excludes it and why a pure sinusoid is not BIBO stable.

2.5 Inverse Z-transform: the three methods

Determine the inverse Z-transform of the given $X(z)$ for the stated ROC 5 6 1+5 2+6

TOP 32/45 · the single biggest question in this chapter, 5 or 6 marks in nearly every paper · every instance is worked in Ch2-num

Which method, and when

1. **Partial fractions:** the default. Use unless the question names another method. Gives a closed form for every n .
2. **Long division (power series):** use when the question says so (77 Ch, 76 Bh, 73 Bh, 72 Ma, 70 Ma). Gives numbers, **not a closed form**, so give 4 or 5 terms and the recurrence.
3. **By inspection:** when $X(z)$ is already a finite polynomial, read the coefficients straight off as impulses (74 Ash, 78 Ch).
4. The contour-integral / residue method is never asked in these 45 papers.

The partial-fraction method, in one fixed order

1. Is $X(z)$ proper? If the numerator's degree is $\geq$ the denominator's, **long-divide first** and keep the polynomial part aside: it becomes impulses.
2. **Divide by z** and expand $X(z)/z$ in partial fractions.
3. Multiply back by z , so every term reads $\frac{Az}{z-a}$.
4. **Assign each pole a side** from the ROC (§2.2).
5. Write the answer term by term with the pairs opposite.

Long division: which way to divide (where the marks are lost)

- **Right-sided** (ROC outside, causal): arrange both polynomials in **ascending powers of z^{-1}** and divide. The quotient $x[0] + x[1]z^{-1} + x[2]z^{-2} + \dots$ is read off directly.
- **Left-sided** (ROC inside, anticausal): arrange in **ascending powers of z** instead. The quotient gives $x[-1]z, x[-2]z^2, \dots$
- **Dividing the wrong way round gives the other sequence**, which is a valid series for a different ROC: and scores zero.

Why divide by z first

- The standard pairs are $\frac{z}{z-a} \leftrightarrow a^n u[n]$, with a z on top.
- Expanding $X(z)$ directly gives terms $\frac{A}{z-a}$ with no z on top, and each then needs an extra time shift. Expanding $X(z)/z$ puts the z back for free.
- **Use it every single time**, including the repeated-pole cases. One idiom, no decisions.

The pair table, both sides

Term	ROC $ z > a $	ROC $ z < a $
$\frac{z}{z-a}$	$a^n u[n]$	$-a^n u[-n-1]$
$\frac{z}{(z-a)^2}$	$n a^{n-1} u[n]$	$-n a^{n-1} u[-n-1]$
constant C	$C \delta[n]$	$C \delta[n]$
z^k	$\delta[n+k]$, an <i>advance</i>	

The right-hand column is just the left with a minus sign and $u[-n-1]$ instead of $u[n]$. That is the only thing the ROC changes.

- Faster than dividing: read the **recurrence** straight off the denominator. For $X(z) = \frac{1}{1 - a_1 z^{-1} - a_2 z^{-2}}$,

$$x[n] = a_1 x[n-1] + a_2 x[n-2], \quad x[0] = 1$$

and then just iterate. This is the same difference equation the filter implements (Ch 4).

- **Always give the recurrence as well as the terms:** it shows the pattern continues and it is what a 6-mark answer wants.

2.6 Last-minute recall for this chapter

Must memorise

- $X(z) = \sum_n x[n]z^{-n}$; the DTFT is $X(z)$ on the unit circle.
- $a^n u[n] \leftrightarrow \frac{z}{z-a}, |z| > |a|$; $-a^n u[-n-1] \leftrightarrow \frac{z}{z-a}, |z| < |a|$.
Same $X(z)$, different ROC.
- $na^n u[n] \leftrightarrow \frac{az^{-1}}{(1-az^{-1})^2}$; $\frac{z}{(z-a)^2} \leftrightarrow na^{n-1}u[n]$.
- The 8 ROC properties (§2.2): **written out, in order.**
- Causal $\Rightarrow$ ROC outside the outermost pole. Stable $\Rightarrow$ ROC contains the unit circle. **Both $\Rightarrow$ all poles inside the unit circle.**
- Convolution $\rightarrow$ product; shift $\rightarrow z^{-k}$; multiply by $a^n \rightarrow X(a^{-1}z)$; multiply by $n \rightarrow -z dX/dz$.
- Inverse: proper? $\rightarrow$ divide by $z \rightarrow$ partial fractions $\rightarrow$ assign sides from the ROC.

Most repeated, in order

1. Inverse Z-transform with a given ROC **TOP 32/45**
 2. Properties of the ROC **TOP 14/45**
 3. Find $X(z)$ and locate its ROC **HOT 6/45**
 4. Convolution property, stated and proved **HOT 5/45**
 5. Z-transform of $na^n u[n]$ **PIN 3/45**
 6. One-sided Z-transform and its time-delay property **PIN 2/45**
 7. Time-shift property, derived **PIN 1/45**
 8. Scaling property with an example **PIN 1/45**
- Q3 of the paper is the ROC properties plus an inverse Z-transform in 30 of the 45 papers.** If you prepare one question in this subject, prepare that one.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full. $\rightarrow$ reads “leads to / therefore”, $+$ reads “and”, $=$ reads “is”. $\checkmark$ means the property holds, $\times$ that it does not.

Short	Means	Short	Means
DSP	digital signal processing	DSAP	digital signal analysis and processing
DT	discrete-time	CT	continuous-time
LTI	linear time-invariant	LSI	linear shift-invariant, the same thing
TI	time-invariant	BIBO	bounded input, bounded output
LCCDE	linear constant-coefficient difference equation	ROC	region of convergence
DTFS	discrete-time Fourier series	DTFT	discrete-time Fourier transform
CTFT	continuous-time Fourier transform	DFS	discrete Fourier series
DFT	discrete Fourier transform	IDFT	inverse DFT
FFT	fast Fourier transform	DIT	decimation in time
DIF	decimation in frequency	MAC	multiply and accumulate
FIR	finite impulse response	IIR	infinite impulse response
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
BLT	bilinear transformation	II	impulse invariance
A/D	analog to digital	D/A	digital to analog
S/H	sample and hold	ZOH	zero-order hold
SNR	signal to noise ratio	PYQ	past-year question
coeff	coefficient	freq	frequency
eqn	equation	no.	number
ops	operations	attr	attribute
w/	with	w/o	without
Eg	for example	Vs	compare / differentiate
+Fig	a figure is wanted	(I)	importance / need
(+)	advantage	(-)	disadvantage

Exam paper codes

Month codes

- **Ba** Baishakh **Jth** Jestha **Asa** Ashad
- **Shr** Shrawan **Bh** Bhadra **Ash** Ashwin
- **Ka** Kartik **Mng** Mangsir **Po** Poush
- **Ma** Magh **Ch** Chaitra
- **Ash and Asa are different months**: 2075 Ashwin and 2070 Ashad are separate papers.

How a year code is written

- **Bold** = **Regular** exam. Plain = Back exam.
 - “New Back (2066 & Later Batch)” counts as Back.
 - A header printing “Regular / Back” in one box counts as Regular.
- Typewriter = the EX 753 paper (DSP, BEX, IV/II).
Normal font = **CT 704** (DSAP, BEI/BCT, IV/I).
- The two axes are independent: **76 Ch** is CT 704 Regular, 76 Ash is CT 704 Back, 76 Bh is EX 753 Regular, 74 Ma is EX 753 Back.
- **69 Bh and 69 Bh are two different papers**, one in each course.

Frequency chips

- **TOP n/45** asked in **10 or more** of the 45 papers.
- **HOT n/45** asked in **4 to 9** papers.
- **PIN n/45** asked in **1 to 3** papers.
- **verified/added** written or corrected against a reference because the source notes did not cover it, or covered it wrongly.
- **5** marks the question has actually carried. Several chips mean the same question has been set at different marks in different years.
- The count is over **45 papers**: 28 CT 704 (2066 Magh to 2082 Bhadra) and 17 EX 753 (2069 Bhadra to 2081 Chaitra).
- Counts come from the Detailed PYQ document, which reproduces every question in full.